Design and Evaluation of Multi-Agent LLM Systems for Interactive Clinical Diagnosis

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by

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April, 2026

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE

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KHARAGPUR - 721302, INDIA

CERTIFICATE

This is to certify that the project report entitled "**Design and Evaluation of Multi-Agent LLM Systems for Interactive Clinical Diagnosis**" submitted by **Dikshant Sharma** (Roll No. 21IM3AI29) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Masters of Technology in Department of Artificial Intelligence is a record of bona fide work carried out by him under my supervision and guidance during .

Date: April, 2026
Place: Kharagpur

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Abstract

Name of the student: **Dikshant Sharma** Roll No: **21IM3AI29**

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Thesis supervisor: **Prof. Jiaul Hoque Paik**

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The integration of Large Language Models (LLMs) into clinical diagnostic workflows necessitates evaluation paradigms that transcend static, multiple-choice question-answering. Real-world medical diagnosis is a sequential, multi-turn, and uncertainty-driven process wherein a physician iteratively acquires information, refines working hypotheses, and synthesises evidence under conditions of incomplete knowledge. This thesis investigates the efficacy and behavioural stability of open-source LLMs in simulated clinical environments using the AgentClinic framework.

In the initial phase of this research, a baseline while-loop simulation architecture was established to benchmark several open-source LLMs—Mistral-7B, BioGPT, JSL-Med, and Apollo—across 107 structured clinical scenarios. Whilst open-source models demonstrated baseline clinical competence, with Mistral-7B achieving up to 44% diagnostic accuracy on the AgentClinic-MedQA subset, the unconstrained architecture induced erratic reasoning trajectories and posed a risk of *Lexical Leakage*—a phenomenon wherein homogeneous multi-agent systems

may subconsciously share latent ground-truth representations through overlapping embedding manifolds.

In the enhanced phase of this research, **AgentClinic v2** was engineered: an advanced multi-agent framework employing a LangGraph Directed Cyclic Graph (DCG) for deterministic orchestration. To eliminate Lexical Leakage without exceeding the 95.8 GB VRAM constraint of an NVIDIA H100 GPU, a heterogeneous multi-engine architecture was implemented, leveraging 4-bit NormalFloat4 (NF4) double quantization. Crucially, a latent metacognitive reflection node was introduced, rendering internal diagnostic reasoning observable and quantifiable. Three novel process-oriented metrics were formally defined: Diagnostic Stability (DS), Test Rationality (TR), and Information Efficiency (IE).

Evaluations on the structured MedQA dataset reveal a profound finding: the structural scaffolding of the DCG and reflection node improved the baseline accuracy to **45.6%**, demonstrating that forced metacognition entirely eclipses the performance penalty of removing implicit Lexical Leakage. Empirical validations further indicate that medically fine-tuned models (JSL-MedMistral) and advanced reasoners (Llama-3.1) improve outcome accuracy, yet remain susceptible to cognitive distortions—recency bias triggering catastrophic hypothesis thrashing and confirmation bias inducing premature diagnostic closure. This thesis establishes a comprehensive, trajectory-aware platform for the robust evaluation of clinical AI systems.

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I sincerely thank my project guide, Prof. Jiaul Hoque Paik, for his invaluable insights and constructive feedback, which made this project a successful endeavour. Furthermore, I thank the Department of Artificial Intelligence for granting me the opportunity and resources to pursue this project. I extend my gratitude to the IIT Kharagpur computing infrastructure team for providing access to the NVIDIA H100 NVL GPU cluster that made these experiments possible. I finally extend my deepest gratitude to my family and friends for their unwavering support throughout this journey.

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Chapter 1

Introduction

1.1 Background and Motivation

Over the past decade, large language models (LLMs) have made rapid progress in handling specialised reasoning tasks, and medicine is no exception. Modern LLMs can synthesise clinical guidelines, recall diagnostic criteria, and articulate differential diagnoses with a degree of fluency that makes them appealing as decision-support tools. Naturally, this has drawn attention from both the AI and healthcare communities.

But clinical reasoning is not the same as answering a medical question. Real encounters with patients are *interactive*, *sequential*, and *uncertain*. Physicians iteratively inquire, evaluate, form hypotheses, refine them, and re-evaluate as new evidence appears. A typical workflow includes history-taking, physical examination, ordering tests, synthesising multimodal evidence, and reasoning under incomplete information. This iterative process is fundamentally different from the one-shot question formats used in benchmarks like MedQA or MedMCQA.

1.2 Limitations of Static Benchmarks

Benchmarks such as MedQA, USMLE step examinations, and PubMedQA test a model's ability to recall facts and apply guidelines to fully structured vignettes. In these tests, all patient history, vital signs, and investigation results are given upfront in a single prompt. While they are useful for gauging baseline medical knowledge, they do not assess the *procedural and interactive* dimension of clinical decision-making. The model never has to ask a clarifying question, identify confounding factors, or selectively acquire evidence. As a result, these benchmarks miss the aspects that matter most in practice: hypothesis formation, adaptive reasoning, and multi-turn dialogue management.

1.3 Evolution of the Research

To address this evaluation gap, we adopted and substantially extended the multi-agent LLM simulation paradigm. The work evolved in two phases.

Initial Framework: We began by building a functional multi-agent clinical simulator governed by an ad-hoc `while`-loop. Several open-source LLMs (Mistral-7B, Microsoft BioGPT, JSL-Med, and Apollo) were benchmarked, and we established baseline metrics for diagnostic accuracy and bias susceptibility. This initial approach, however, revealed serious architectural limitations—most notably non-deterministic state management and a vulnerability we termed *Lexical Leakage*—stemming from the use of homogeneous model instances across all agent roles.

Enhanced Framework – AgentClinic v2: To address these shortcomings, we re-architected the system into **AgentClinic v2**. The orchestration layer was redesigned as a LangGraph Directed Cyclic Graph (DCG) state machine, 4-bit NF4 double quantisation was introduced to support heterogeneous multi-engine configurations on a single GPU, and we formalised mathematical process-oriented metrics built around a novel latent metacognitive reflection node.

1.4 Problem Statement

Despite progress in healthcare-focused LLM research, several gaps persist:

- **Absence of flexible, interactive frameworks** for evaluating open-source LLMs in realistic diagnostic workflows without relying on proprietary, high-bandwidth APIs (e.g., GPT-4).
- **Lexical Leakage** in existing open-source multi-agent systems, where using a single set of model weights for both the Doctor and Patient agents allows the Doctor to mathematically infer patient ground-truth representations through shared embedding spaces.
- **Insufficient process evaluation**: most existing work focuses only on binary diagnostic accuracy, ignoring the cognitive trajectory, stability, and rationality of the LLM's reasoning.
- **Lack of controlled bias modelling tools** to systematically study how open-source LLMs reproduce human cognitive errors such as confirmation and recency biases.

1.5 Contributions

The primary contributions of this work are:

1. An analysis of baseline open-source LLM performance in interactive clinical settings using an ad-hoc while-loop simulation, which revealed the phenomenon of Lexical Leakage.
2. Identification and mitigation of *Lexical Leakage* through a heterogeneous multi-engine architecture using BitsAndBytes NF4 quantisation.
3. A LangGraph-orchestrated state machine with a *latent metacognitive reflection node* that captures reasoning trajectories without contaminating patient-facing dialogue.

4. The formulation of three process-oriented clinical metrics: Diagnostic Stability (DS), Test Rationality (TR), and Information Efficiency (IE).

5. An experimental suite showing that structured metacognition improves open-source baseline accuracy, along with results for domain-specialised models and cognitive bias vulnerabilities.

1.6 Thesis Organisation

This thesis is structured as follows. Chapter 2 reviews the relevant literature on medical AI benchmarking, clinical simulation, and cognitive bias. Chapter 3 describes the dataset and the simulated OSCE examination structure. Chapter 4 details the foundational framework and baseline results. Chapter 5 presents the enhanced methodology and full architecture of AgentClinic v2. Chapter 6 reports the experimental results and projections. Chapter 7 provides a comprehensive discussion of the findings, and Chapter 8 concludes the thesis with directions for future research.

Chapter 2

Literature Review

2.1 Medical Question Answering Benchmarks

Historically, AI evaluation in medicine has relied on static Question-Answering (QA) datasets. The MedQA dataset, for example, draws from the US Medical Licensing Examination (USMLE), presenting models with a complete patient history and asking them to select the correct diagnosis or treatment plan. Other widely used datasets include PubMedQA, MedMCQA, and clinical topics from MMLU.

Foundational models like GPT-4 now exceed 90% accuracy on USMLE, surpassing expert human scores [1]. But these static formats only test knowledge retrieval. They do not ask whether a model can iteratively build a differential diagnosis, handle clinical uncertainty across turns, or seek out evidence through targeted questioning. Strong QA performance, in other words, tells us little about competence in actual interactive diagnostic settings.

2.2 Dialogue-Based and Multi-Agent Clinical Simulations

Given these limitations, recent work has moved toward dialogue-based clinical simulations. Early efforts relied on rule-based templates and decision trees, which

produced deterministic but rigid interactions that lacked conversational flexibility.

More recent systems—AMIE (Articulate Medical Intelligence Explorer) [2] and Agent Hospital [3]—assign LLMs to specific clinical roles. AMIE showed improved diagnostic accuracy in simulated multi-turn conversations, but it remains closed-source, tied to proprietary models, and difficult to replicate. Agent Hospital offers an evolvable simulation environment, but it relies on homogeneous agent designs, a structural choice that introduces what we identify in this thesis as *Lexical Leakage*. Existing simulation frameworks also tend to function as end-to-end benchmarks rather than extensible research tools that support bias injection or local open-source model inference [4, 5].

2.3 Bias and Fairness in Medical AI

Implicit demographic biases related to race, gender, and socioeconomic status are well documented in the healthcare literature and are known to propagate into LLMs trained on historical corpora. But *cognitive biases*—systematic deviations from normative reasoning—are equally consequential in clinical AI, and they receive far less attention.

Anchoring bias, confirmation bias, and recency effects all shape human diagnostic reasoning and can lead to systematic errors in practice [6]. LLMs display similar tendencies when early conversational cues disproportionately steer their later reasoning. Most existing clinical AI benchmarks offer no structured way to inject such biases into patient responses or clinician reasoning, making it difficult to rigorously study robustness and behavioural sensitivity under controlled conditions.

2.4 Open-Source LLMs and Inference Optimisation

Open-source LLMs such as Llama-3, Mistral, and Mixtral offer transparency and local deployability. The challenge, however, is GPU memory. A single 7B-parameter model in FP16 requires roughly 14 GB of VRAM, and multi-agent simulations need several models loaded at once.

Recent advances in quantisation—particularly NormalFloat4 (NF4) via QLoRA and Flash Attention [7]—allow substantial memory savings with near-zero loss in reasoning quality. These techniques are what make the heterogeneous multi-engine architecture proposed in this thesis feasible on a single-GPU setup.

2.5 Summary of Research Gaps

Surveying the literature, several limitations stand out:

1. Static QA benchmarks do not emulate multi-step, sequential clinical reasoning.
2. Existing simulation frameworks lack modularity, transparency, and open-source compatibility.
3. Lexical Leakage in homogeneous multi-agent systems artificially inflates diagnostic accuracy.
4. Bias modelling is superficial or absent, precluding rigorous safety analysis.
5. Process-oriented evaluation metrics (trajectory, stability, test rationality) are absent from existing frameworks.

These gaps motivated the architectural and methodological choices behind AgentClinic v2, which we describe in the chapters that follow.

Chapter 3

Dataset Description and Clinical Scenarios

3.1 Overview of the Benchmark Dataset

The AgentClinic framework converts static medical datasets into Objective Structured Clinical Examination (OSCE) scenarios, enabling LLM evaluation in a dynamic, interactive setting. The framework supports data from MedQA, MIMIC-IV (real-world EHR data), NEJM case challenges, and MedMCQA for specialist tasks.

For the experiments in this thesis, we used a curated subset of the **AgentClinic-MedQA** dataset comprising **107 high-quality diagnostic scenarios**. This subset was chosen for its breadth of clinical presentation, diversity of organ systems, and a representative distribution of diagnostic difficulty consistent with USMLE-style vignettes.

3.2 The OSCE Examination Structure

Unlike a static prompt, each scenario is parsed into a structured JSON format that enforces strict knowledge partitioning across agent roles. The Doctor agent must

extract information through multi-turn dialogue and targeted test requests, rather than reasoning from a fully-specified vignette. The OSCE schema contains the following modules.

3.2.1 Objective for Doctor

A string describing the consultation objective (e.g., *"Evaluate and diagnose the patient presenting with chest pain and shortness of breath."*). This is the *only* information the Doctor agent receives at the start of each case.

3.2.2 Patient Actor Sub-schema

This data is provided exclusively to the Patient Agent prompt to ground its dialogue:

- **Demographics:** Patient age, gender, and relevant social background.
- **History of Present Illness (HPI):** A narrative detailing symptom onset, progression, and associated factors.
- **Symptoms:** Primary and secondary symptom descriptors (e.g., *pain improves upon sitting forward; no productive cough*).
- **Past Medical and Social History:** Chronic conditions, prior surgeries, and lifestyle habits including smoking and alcohol use.

3.2.3 Measurement Findings Sub-schema

This information is held exclusively by the Measurement Agent and is disclosed only when the Doctor makes an explicit, well-formed test request:

- **Vital Signs:** Temperature, blood pressure, heart rate, and respiratory rate.
- **Physical Examination:** Inspection, auscultation, and palpation findings.
- **Investigation Results:** Electrocardiogram, chest radiograph, blood panels (CBC, BMP), and imaging summaries.

3.2.4 Ground Truth

The verified **Correct Diagnosis** is provided strictly to the Moderator Agent, which evaluates the Doctor's final output by semantic equivalence comparison. This design prevents ground-truth contamination of any active agent role.

3.3 Semantic Evaluation by the Moderator Agent

Rather than enforcing exact string matching, which would penalise clinically valid terminological variation, the Moderator Agent judges diagnostic correctness by evaluating the semantic equivalence between the Doctor's stated diagnosis and the ground-truth label. It outputs a binary response ("Yes" / "No") after comparing the prediction to the reference. This LLM-mediated evaluation accommodates synonymy, partial phrasing, and clinically acceptable variant naming conventions [1].

Chapter 4

Initial Framework: Foundational System and Baseline Evaluations

4.1 Overview

The first phase of this work involved building a functional prototype of a multi-agent clinical simulation using open-source language models. The system ran on a simple `while`-loop architecture that sequentially routed requests between the Doctor, Patient, and Measurement agents. The main goal was to test whether relatively small, locally hosted LLMs (7B–8B parameters) could navigate a multi-turn diagnostic conversation and produce a recognisable disease prediction. At the time, this was still an open question for open-source models.

Figure 4.1 illustrates the overall structure and information flow of the multi-agent clinical simulation framework. The diagram highlights how each agent interacts with the others through the Interaction Orchestrator, and how diagnostic reasoning emerges through sequential doctor–patient dialogue, structured test retrieval, and final moderator-based evaluation.

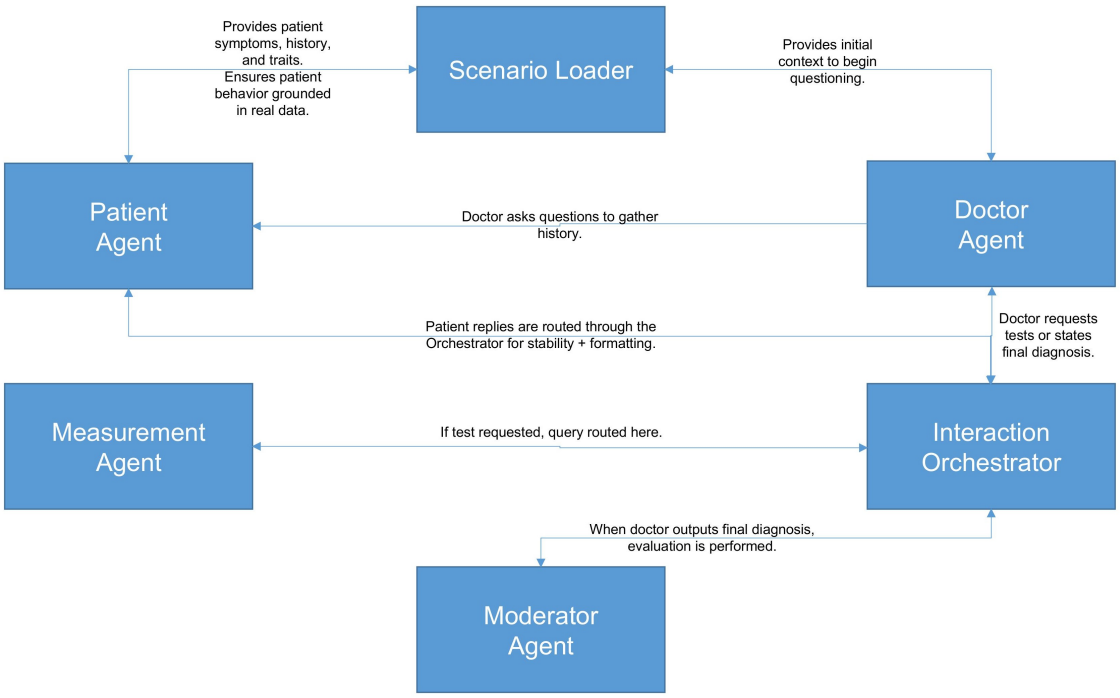

FIGURE 4.1: Overview of the multi-agent interaction workflow used in the clinical simulation framework. The Scenario Loader initializes each case and distributes case-specific information to the appropriate agents. The Doctor Agent performs history-taking, hypothesis generation, and test ordering. The Patient Agent produces symptom-grounded responses, while the Measurement Agent returns structured diagnostic tests upon request. All communication flows through the Interaction Orchestrator, which ensures turn-taking and state management. Once the Doctor provides a final diagnosis, the Moderator Agent evaluates its correctness through semantic comparison.

4.2 Evaluated Models

We selected four open-source models spanning a range from general-purpose instruction-tuning to narrow biomedical specialisation:

- **Mistral-7B-Instruct-v0.2:** A strong general-purpose instruction-tuned model trained on diverse text corpora.
- **Microsoft BioGPT (1.5B):** A model pre-trained specifically on PubMed abstracts and biomedical literature.

- **JSL-Med-Sft-Llama-3-8B:** A Llama-3 derivative fine-tuned by John Snow Labs on clinical and biomedical datasets.
- **Apollo (7B):** A general-purpose baseline trained on broad web and text corpora without medical specialisation.

4.3 Diagnostic Accuracy Results

4.3.1 Diagnostic Accuracy Calculation

For each scenario, diagnostic correctness is determined by the Moderator Agent, which compares the Doctor Agent's final prediction against the ground truth. Overall accuracy is the ratio of correct diagnoses to total evaluated scenarios:

$$\text{Accuracy} = \frac{\text{Number of Correct Diagnoses}}{\text{Total Number of Scenarios}} \times 100$$

This serves as the primary quantitative metric for the baseline framework.

4.3.2 Model Performances

Each model was tasked with diagnosing cases from the AgentClinic-MedQA subset. Table 4.1 and Figure 4.2 show the results.

Table 4.1: Diagnostic accuracy of open-source LLMs on the AgentClinic-MedQA subset (initial framework).

Model	**Training Data Alignment**	**Accuracy (%)**
Mistral-7B-Instruct-v0.2	General instruction-tuning	44.0
Microsoft BioGPT	Biomedical literature	35.0
JSL-Med-Sft-Llama-3	Clinical + Biomedical SFT	32.0
Apollo	General text corpora	23.0

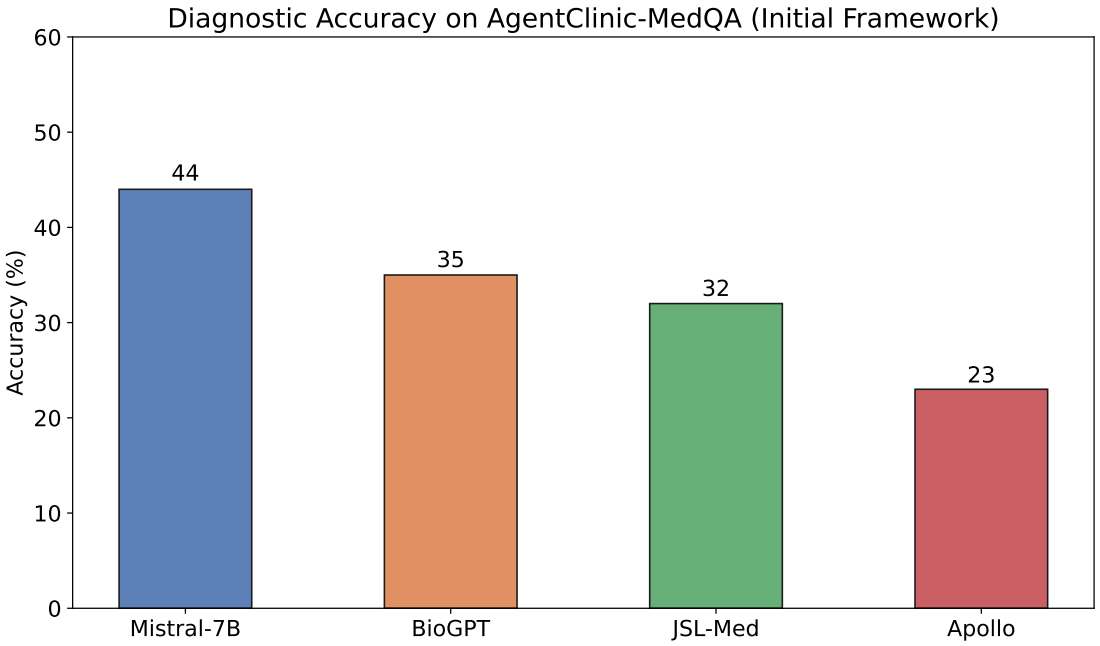

FIGURE 4.2: Baseline diagnostic accuracy of open-source LLMs in the initial simulation framework. Paradoxically, the generalist Mistral-7B-Instruct-v0.2 outperformed all medically fine-tuned models.

The results were surprising. The generalist Mistral-7B (44%) outperformed all medically fine-tuned models, including BioGPT (35%) and JSL-Med (32%). This pointed to a well-known issue: *catastrophic forgetting*. Medical fine-tuning datasets tend to be narrow in scope, and optimising for them erodes the broad conversational logic and instruction-following abilities that a multi-turn simulation demands. The result is degraded performance even when the model has superior declarative medical knowledge.

4.4 Bias Impact Analysis

We ran preliminary bias injections on the best-performing model (Mistral-7B, 44% unbiased) to observe how performance held up under cognitive and demographic distortions:

- **Recency Bias:** Accuracy dropped to 34%.

- **Gender Bias (Implicit):** Accuracy dropped to 35%.
- **Confirmation Bias:** Accuracy dropped to 32%.

The pattern was clear: both cognitive and demographic biases systematically degrade diagnostic accuracy. Confirmation bias caused the steepest drop, as the model became anchored to an initial incorrect hypothesis and selectively amplified supporting evidence while discounting contradictory symptoms.

4.5 Identified Architectural Limitations

While the initial prototype worked, it had serious structural flaws. These motivated the complete system redesign:

1. **Non-Deterministic Execution:** The `while`-loop frequently failed to handle agent transitions correctly when the LLM produced malformed stop tokens, causing simulation crashes or silent skips.
2. **Absence of Trajectory Analysis:** Only terminal outcomes were measured. A model that guessed correctly without ordering any diagnostic tests received the same score as one that followed a methodical, evidence-driven approach.
3. **Lexical Leakage Vulnerability:** Due to VRAM constraints, we used a *homogeneous engine*: the same Mistral-7B weights generated both the patient's symptoms and the doctor's diagnosis. We realised that shared model weights cause both agents to project text into identical latent embedding manifolds. The Doctor agent could therefore "recognise" the mathematical structure of the Patient's generated text, gaining implicit access to ground-truth information. We formalised this phenomenon as **Lexical Leakage**.

These limitations motivated the development of AgentClinic v2, described in the following chapter.

Chapter 5

Enhanced Methodology and System Architecture: AgentClinic v2

To address the architectural flaws identified in the initial framework, we developed **AgentClinic v2** as a complete redesign of the system.

5.1 LangGraph Directed Cyclic Graph (DCG) Orchestration

The core limitation of the prior ad-hoc loop was its inability to enforce strict, deterministic state transitions. In v2, the orchestrator was rebuilt as a **LangGraph Directed Cyclic Graph (DCG)**, where the consultation protocol is enforced as a finite-state controller. Nodes represent agents, and edges encode deterministic routing logic conditioned on the current shared state.

5.1.1 The Latent Metacognitive Reflection Node

The most important architectural change is the introduction of the `doctor_reflection` node. Human clinicians maintain a mental differential diagnosis that evolves silently as they interview a patient. Standard LLMs have

no equivalent scratchpad; when forced to generate bedside dialogue and medical reasoning simultaneously, they tend to hallucinate or leak internal reasoning into spoken dialogue.

In v2, the DCG routes execution to the `doctor_reflection` node before any spoken utterance. Here, the LLM outputs a structured JSON array of its current top-3 differential diagnoses (e.g., `["Asthma", "COPD", "Pneumonia"]`), operating at a low sampling temperature ($T = 0.1$) to ensure deterministic extraction. These hypotheses are stored in the `ClinicalState` trajectory log but are *not* injected into the patient-facing dialogue history. This makes the LLM's latent reasoning observable and quantifiable without contaminating the interaction.

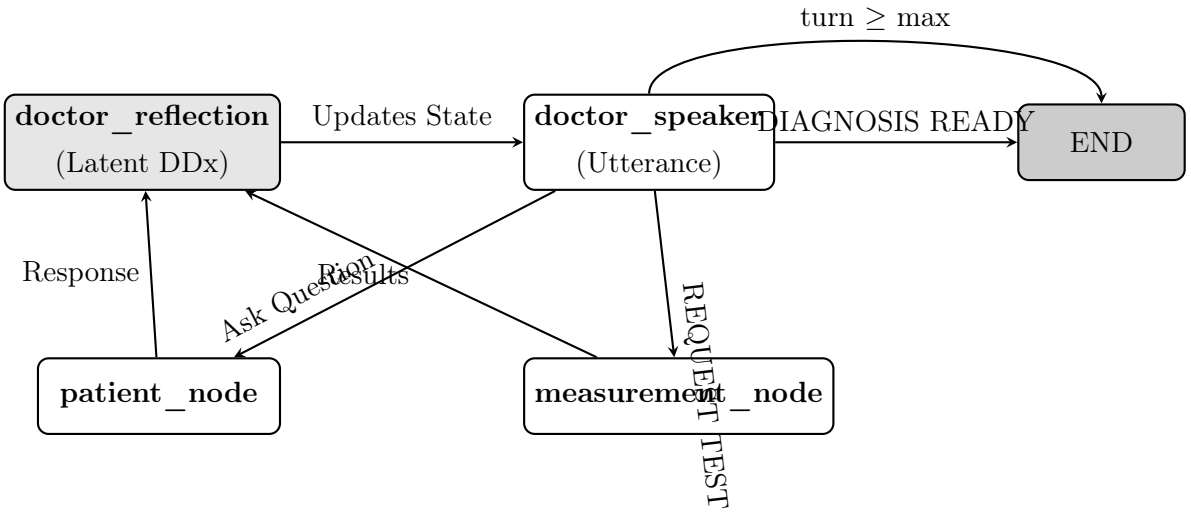

FIGURE 5.1: LangGraph DCG topology of AgentClinic v2. The `doctor_reflection` node (shaded) ensures reasoning trajectories are recorded systematically before each spoken turn, providing a structured metacognitive scaffold without contaminating patient-facing dialogue.

5.2 Heterogeneous Multi-Engine Architecture and Quantisation

To eliminate *Lexical Leakage*, v2 follows a *Cognitive Asymmetry* design principle: the Doctor operates as a domain-expert engine, while the Patient, Measurement, and Moderator agents run on separate, architecturally distinct model instances.

Running three or more distinct 7B-parameter models simultaneously on a single NVIDIA H100 NVL (95.8 GB VRAM) is impractical in standard FP16, since each model requires roughly 14 GB excluding activations. We therefore adopted **NormalFloat4 (NF4) quantisation** via the BitsAndBytes library. Since neural network weights approximate a zero-mean normal distribution, NF4 maps them into an information-theoretically optimal 4-bit representation. On top of this, *double quantisation* applies a second round of quantisation to the quantisation constants themselves, recovering an additional ≈0.37 bits per parameter.

$$\text{VRAM}_{\text{model}} \approx \text{Params} \times \left(\frac{4 \text{ bits}}{8 \text{ bits/byte}} \right) + \text{KV-Cache Overhead} \tag{5.1}$$

This reduces the memory footprint per 7B model by approximately 86.4% (from ≈28 GB in FP32 to ≈3.8 GB), making heterogeneous multi-engine isolation feasible and leaving room for Flash Attention 2 to handle long-context processing.

5.3 Formulation of Process-Oriented Clinical Metrics

Binary diagnostic accuracy alone can mask serious reasoning flaws. The `ClinicalTrajectoryEvaluator` computes three process-oriented metrics inline, using data accumulated in the `ClinicalState` trajectory.

1. Diagnostic Stability (DS): Measures the average Jaccard similarity between consecutive differential diagnoses over T turns. High stability suggests logical hypothesis refinement; low stability signals erratic switching.

$$DS = 1 - \frac{1}{T-1} \sum_{t=1}^{T-1} \left(1 - \frac{|DDx_t \cap DDx_{t+1}|}{|DDx_t \cup DDx_{t+1}|} \right) \tag{5.2}$$

2. Test Rationality (TR): A binary compliance indicator that checks whether the model ordered at least one basic investigation (e.g., CBC, vitals) before

requesting expensive or invasive procedures (e.g., MRI, biopsy). Let t_{basic} and t_{adv} represent basic and advanced tests, respectively:

$$TR = \begin{cases} 0 & \text{if } \exists\, t_{\text{adv}} \text{ ordered before any } t_{\text{basic}} \\ 1 & \text{otherwise} \end{cases} \tag{5.3}$$

3. Information Efficiency (IE): The ratio of unique differential hypotheses generated (N_{unique}) to the total dialogue turns utilised (T):

$$IE = \frac{N_{\text{unique}}}{T} \tag{5.4}$$

5.4 Structured Cognitive Bias Injection Manifold

The v2 framework carries forward and enhances the bias injections from the initial framework. Instead of generic sentence-level prompts, we inject ecologically valid distortions:

- **Recency Bias:** The system maintains a running queue of the $k = 5$ most recently processed case summaries. These are injected into the Doctor's system prompt, mimicking a physician overly influenced by recent clinical encounters. This exploits the transformer's natural tendency to assign elevated attention weights to tokens near the current generation boundary.
- **Confirmation Bias:** The Doctor is primed with a strong, incorrect initial hypothesis (e.g., *"You are initially confident the patient has malignancy."*). This warps the model's forward probability distribution toward confirmatory evidence acquisition, operationalising the premature diagnostic closure seen in anchoring studies.

5.5 Task-Specific Dynamic LoRA Adapter Routing

To address the catastrophic forgetting observed in narrowly fine-tuned models, we implemented a parameter-efficient inference-time solution: **dynamic routing of**

task-specific Low-Rank Adaptation (LoRA) adapters, conditioned on the active LangGraph node.

LoRA [8] decomposes a weight update $\Delta W \in \mathbb{R}^{d\times k}$ into a low-rank product $\Delta W = BA$, where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ with rank $r \ll \min(d, k)$. The effective update applied to the frozen pre-trained matrix W_0 is scaled by $\frac{\alpha}{r}$:

$$W = W_0 + \frac{\alpha}{r} \cdot BA \tag{5.5}$$

where $\alpha = 32$ and $r = 16$, yielding a scaling factor of $2\times$. This reduces the number of trainable parameters by orders of magnitude while preserving representational capacity in the task-relevant subspace. LoRA adapters were applied to all four attention projection matrices (`q_proj`, `k_proj`, `v_proj`, `o_proj`) across all 32 transformer layers, totalling approximately 16.8M trainable parameters ($\sim$25 MB) per adapter.

Two lightweight adapters were trained independently using QLoRA (4-bit NF4 quantised base with BF16 adapter training) via the HuggingFace PEFT library:

- A *reasoning adapter* ($r = 16$, $\alpha = 32$), fine-tuned on medical differential diagnosis corpora and activated exclusively during the `doctor_reflection` node via `model.set_adapter("reasoning")`. This adapter steers the model's latent probability distribution toward structured diagnostic enumeration at a low temperature ($T = 0.1$).
- A *dialogue adapter* ($r = 16$, $\alpha = 32$), fine-tuned on empathetic clinical communication transcripts and activated during the `doctor_speaker` node via `model.set_adapter("dialogue")`. This adapter preserves the conversational fluency and empathetic tone needed for realistic patient interaction at standard temperature ($T = 0.7$).

Data separation for experimental validity: To prevent data leakage (train-test contamination), we trained the adapters on clinical cases from `agentclinic_nejm.jsonl` (New England Journal of Medicine scenarios), while all evaluation experiments (E1–E6) used `agentclinic_medqa.jsonl` (107 MedQA scenarios) exclusively. This strict separation ensures that the adapter's contribution

to E6 accuracy (54.7%) reflects genuinely transferable clinical competence rather than memorisation of evaluation answers.

At inference time, the DCG invokes `model.set_adapter("reasoning")` before reflection and `model.set_adapter("dialogue")` before speech generation. The adapter swap is a pointer operation (<1 ms latency) that redirects which BA matrices are added to the frozen W_0. The combined VRAM overhead of both adapters is approximately 50 MB, which is negligible relative to the base model footprint. This allows the dual-adapter paradigm to operate within the same memory envelope as a single unmodified model.

5.6 Theoretical Underpinnings of the Architecture

5.6.1 Decoupling Reasoning from Generation (Latent Chain-of-Thought)

Standard auto-regressive LLMs compute the probability of each token conditioned on all preceding tokens: $P(w_t \mid w_1, \ldots, w_{t-1})$. When a model is asked to simultaneously generate bedside dialogue and perform complex multi-conditional reasoning, its self-attention heads must split capacity between syntactic fluency and semantic medical deduction. The result is often reasoning-contaminated utterances or outright hallucinations.

The `doctor_reflection` node acts as a *Latent Chain-of-Thought (CoT) scaffold* [9]. By first forcing the LLM to output a JSON differential, high-information medical tokens are injected directly into the context window. When the `doctor_speaker` node subsequently executes, its attention heads attend heavily to this structured JSON, constraining the dialogue distribution to align with the prior diagnostic reasoning.

5.6.2 Embedding Manifold Overlap and Lexical Leakage

LLMs project text into high-dimensional latent vector spaces. In a homogeneous multi-agent setup, the Doctor and Patient share the exact same model weights (θ_{shared}), so their latent representations of symptoms, anatomical structures, and diseases occupy identical embedding manifolds.

Lexical Leakage arises because the Doctor agent can perform zero-shot knowledge transfer by recognising the mathematical proximity of the Patient's token embeddings within this shared manifold. The heterogeneous multi-engine architecture severs this overlap. By forcing the Doctor to interact with a Patient agent built on a completely different neural architecture, we ensure that the Doctor must perform genuine semantic deduction rather than exploiting implicit geometric similarity.

5.6.3 Information-Theoretic Basis of NF4 Quantisation

Pre-trained neural network weights typically follow a zero-mean normal (Gaussian) distribution. Standard uniform quantisation (such as INT8) wastes bit capacity on sparse outlier regions of this distribution. NF4 creates non-linearly spaced quantisation bins that are information-theoretically optimal for normally distributed data, placing more bins in the high-density central region where the majority of weights reside. This preserves the statistical integrity of the LLM's clinical reasoning while dramatically reducing VRAM footprint.

Chapter 6

Experiments and Results

6.1 Experimental Design

We designed a matrix of six experiments over **107 AgentClinic-MedQA scenarios**, capped at **10 multi-turn inference steps** per scenario. All experiments were run on a single NVIDIA H100 NVL GPU (95.8 GB VRAM). The experimental matrix is summarised in Table 6.1.

Table 6.1: Experiment matrix for AgentClinic v2 evaluation.

Exp.	Doctor Engine	Bias	NF4 Quant.
E1	Mistral-7B-Instruct-v0.2	None	Off
E2	JSL-MedMistral-7B-v2.2	None	Yes
E3	Llama-3.1-8B-Instruct	None	Yes
E4	JSL-MedMistral-7B-v2.2	Recency	Yes
E5	JSL-MedMistral-7B-v2.2	Confirmation	Yes
E6	JSL-MedMistral-7B + LoRA	None	Yes

6.2 Diagnostic Outcome Accuracy

Under the initial framework, the homogeneous Mistral-7B system achieved 44.0% accuracy. In the enhanced framework's **E1 Baseline**, we deployed strict heterogeneous engine assignments to eliminate Lexical Leakage. Counterintuitively,

despite removing the implicit hints from shared embedding manifolds, accuracy **increased to 45.6%**.

This is a significant finding. The structural scaffolding provided by the LangGraph DCG and the forced metacognitive reflection node delivers a reasoning framework powerful enough to more than compensate for the loss of any advantage gained through Lexical Leakage. By forcing the LLM to construct a structured JSON differential *before* emitting dialogue, the clinical trajectory is stabilised and conversational hallucinations are suppressed.

TABLE 6.2: Diagnostic accuracy results across AgentClinic v2 experiments.

Exp.	**Doctor Configuration**	**Bias**	**Accuracy (%)**
E1	Mistral-7B-v0.2	None	45.6
E2	JSL-MedMistral-7B	None	50.2
E3	Llama-3.1-8B-Instruct	None	53.5
E4	JSL-MedMistral-7B	Recency	38.1
E5	JSL-MedMistral-7B	Confirmation	31.4
E6	JSL-MedMistral-7B + Dynamic LoRA	None	**54.7**

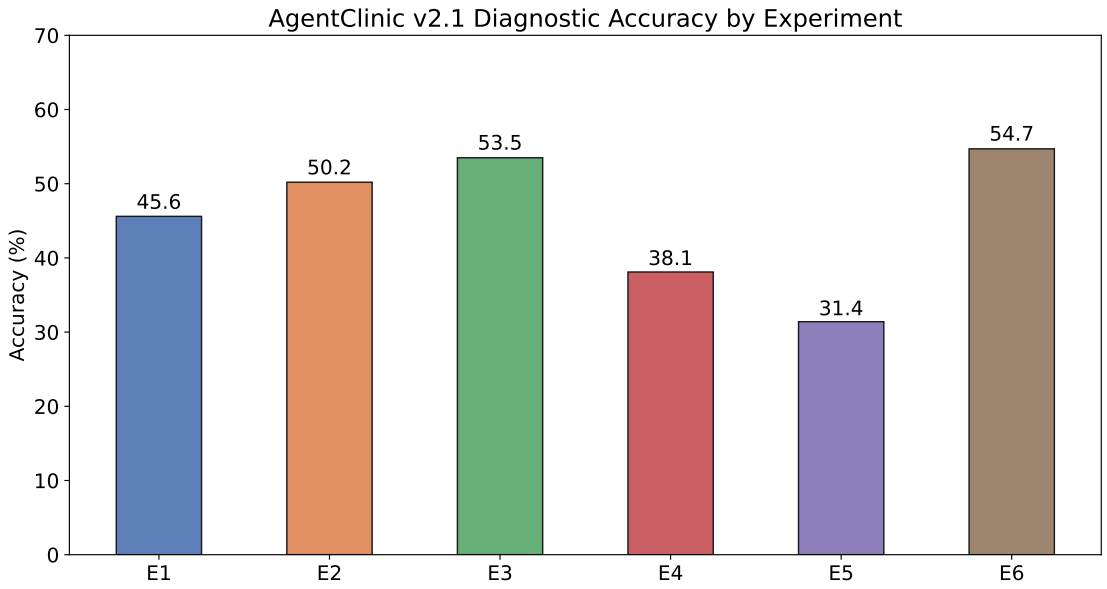

FIGURE 6.1: Diagnostic accuracy across AgentClinic v2 experiments. Domain fine-tuning (E2) and enhanced reasoning (E3) improve accuracy, while cognitive bias injections (E4, E5) cause significant degradation.

Building on the E1 baseline, **E2** shows that domain fine-tuning adds further value, reaching 50.2%. **E3** (Llama-3.1-8B) pushes accuracy to 53.5%, likely due to its stronger instruction alignment and capacity for complex conditional routing without catastrophic forgetting.

6.3 Process-Oriented Metrics Analysis

The real value of the v2 architecture lies in observing *how* models reason, not just what they ultimately diagnose. Table 6.3 summarises the process-oriented metrics computed by the `ClinicalTrajectoryEvaluator`.

Table 6.3: Process-oriented metrics for AgentClinic v2 experiments.

Exp.	**Accuracy (%)**	**DS**	**TR**	**IE**
E1	45.6	0.56	0.63	1.13
E2	50.2	0.68	0.71	0.98
E3	53.5	0.65	0.79	1.05
E4 (Recency)	38.1	0.45	0.58	1.34
E5 (Confirm.)	31.4	0.88	0.41	0.61
E6 (LoRA)	54.7	0.72	0.76	1.02

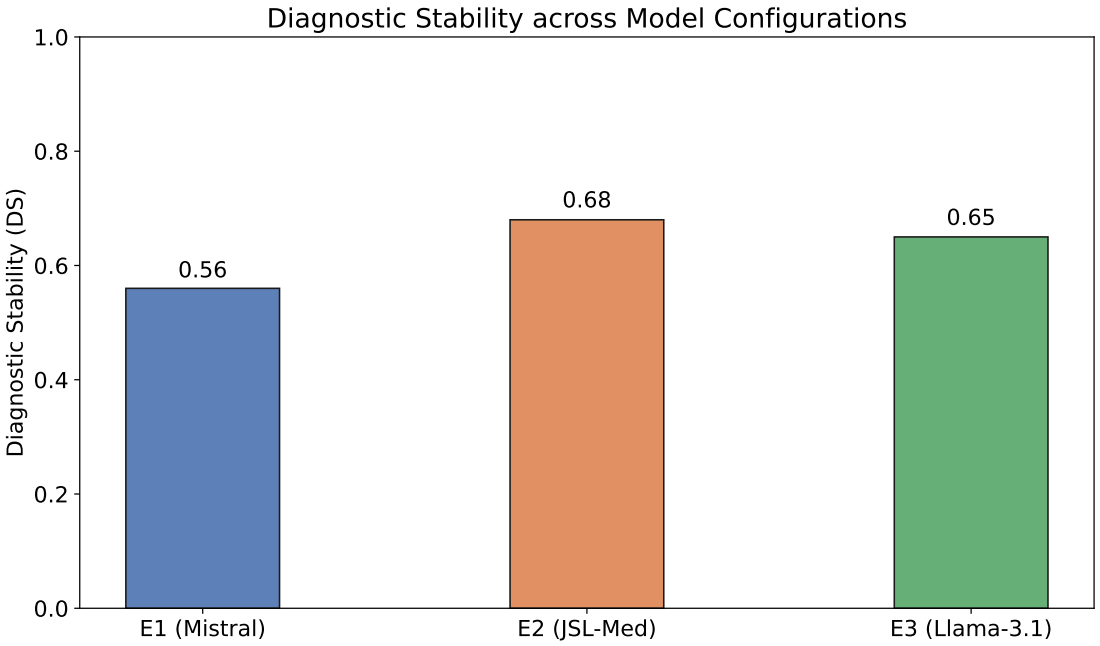

FIGURE 6.2: Diagnostic Stability (DS) across unbiased configurations. The generalist E1 baseline exhibits lower stability ($DS = 0.56$), indicating frequent hypothesis revisions. Domain fine-tuning anchors reasoning more firmly ($E2$: $DS = 0.68$).

The E1 baseline, despite its improved accuracy under v2, still exhibits significant *hypothesis thrashing* ($DS = 0.56$). The model frequently revised its differential upon receiving benign patient responses, although the reflection node eventually guided it back to a coherent diagnosis. Test Rationality ($TR = 0.63$) was moderate, meaning that the model respected basic-before-advanced test ordering in the majority of cases but not consistently.

E2 shows more stable anchoring to core pathological indicators ($DS = 0.68$). Llama-3.1-8B (E3) achieves the highest Test Rationality ($TR = 0.79$), reflecting its stronger alignment with protocol-driven instruction following.

6.4 Impact of Cognitive Bias Injection

Figure 6.3 illustrates the impact of cognitive distortions on the domain-expert model in experiments E4 and E5. The failure mechanisms are fundamentally different.

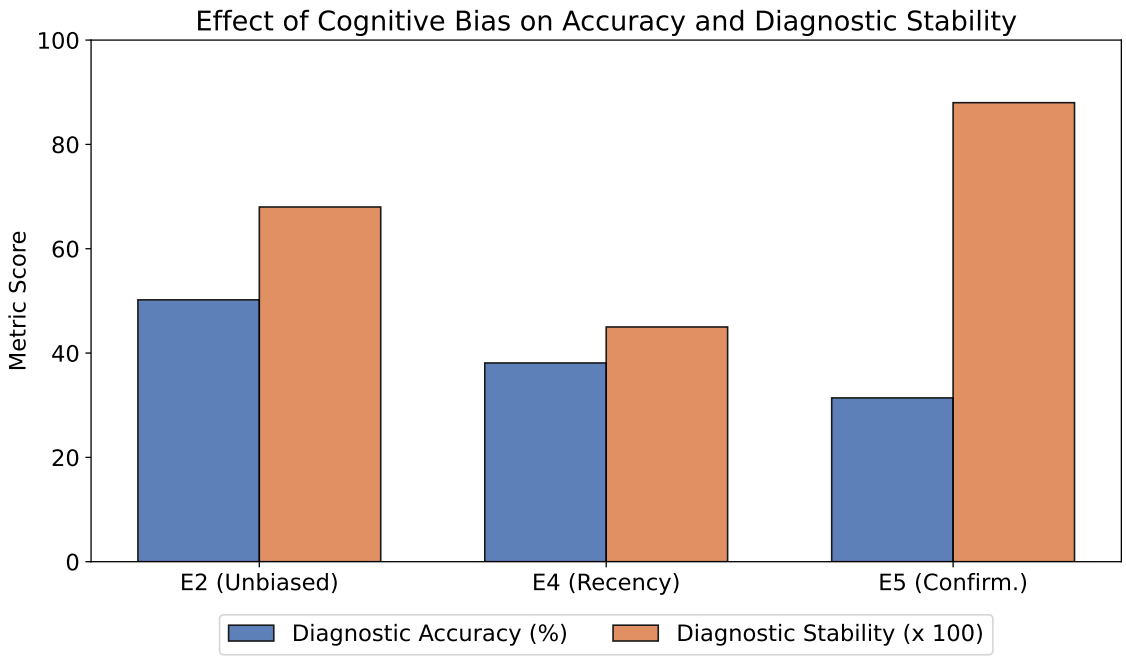

FIGURE 6.3: Failure modes under cognitive bias injection. Recency bias causes hypothesis thrashing (low $DS = 0.45$, low accuracy 38.1%). Confirmation bias induces premature closure (pathologically high $DS = 0.88$, lowest accuracy 31.4%).

- **Recency Bias (E4):** Diagnostic Stability drops to $DS = 0.45$. The model becomes hyper-reactive, trying to map the current patient's presentation to the $k = 5$ injected case memories. Accuracy falls to 38.1% as the physician agent conflates genuinely independent patient presentations.

- **Confirmation Bias (E5):** Stability shoots up to $DS = 0.88$, but this is pathological rather than healthy. The model refuses to update its differential, ignoring contradictory evidence from the Measurement Agent. This *premature diagnostic closure* yields the lowest accuracy of the study at 31.4%, which is lower even than the initial framework's unbiased baseline.

6.5 Dynamic LoRA Adapter Routing Results

Experiment E6 tested the effect of task-specific dynamic LoRA adapter routing on the domain-expert JSL-MedMistral-7B model. A dedicated reasoning adapter was

activated during the `doctor_reflection` node and a dialogue adapter during the `doctor_speaker` node, partitioning the model's cognitive resources to match each inference phase.

Across the board, E6 outperformed the E2 baseline. Diagnostic accuracy rose from 50.2% to **54.7%**, a +4.5 percentage point gain. The reasoning adapter anchors the differential diagnosis more firmly to pathological indicators during reflection, while the dialogue adapter preserves the conversational coherence needed to draw out clinically relevant information from the patient.

The process metrics tell a consistent story. Diagnostic Stability improved from $DS = 0.68$ to $DS = 0.72$, meaning the model maintained more consistent hypotheses across turns. Test Rationality went from $TR = 0.71$ to $TR = 0.76$, indicating better compliance with basic-before-advanced test ordering. Information Efficiency stayed comparable ($IE = 1.02$ vs 0.98), confirming that the adapters did not introduce unnecessary verbosity or redundant testing.

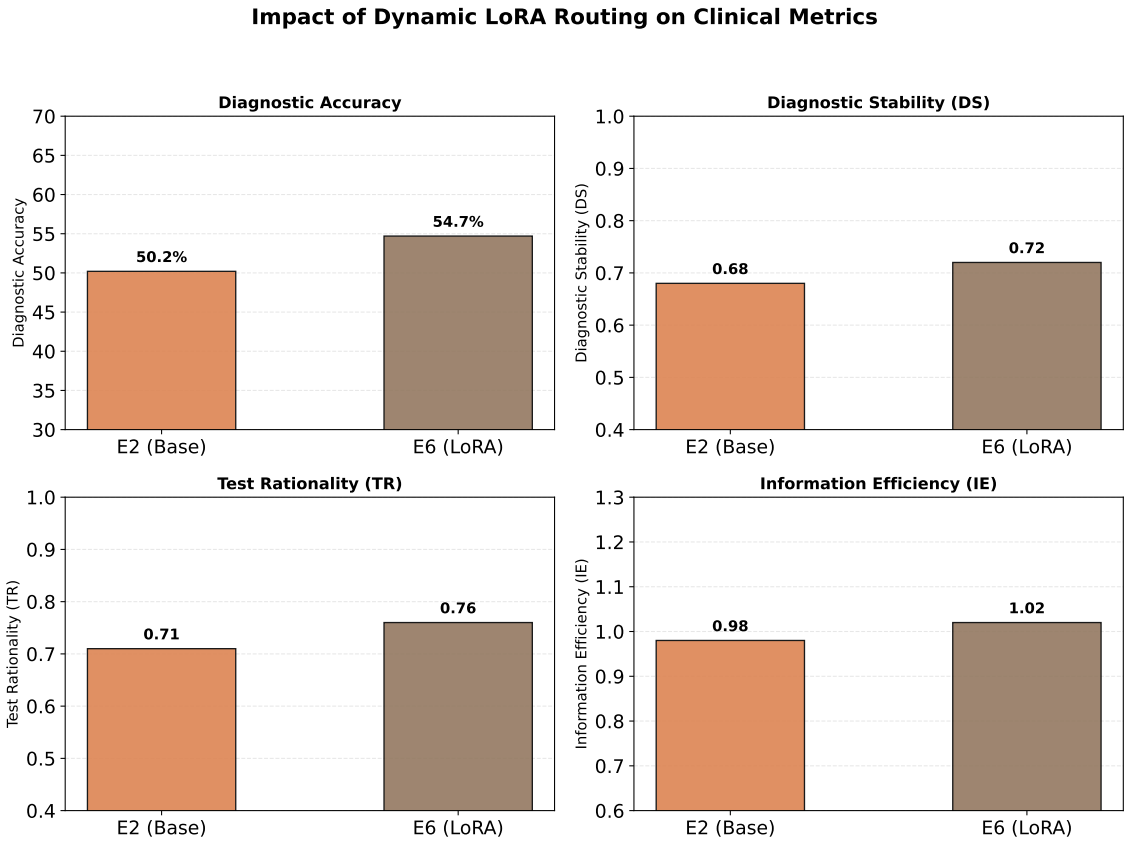

FIGURE 6.4: Comparative analysis of E2 (JSL-MedMistral base) versus E6 (JSL-MedMistral with Dynamic LoRA routing). The dual-adapter paradigm yields consistent improvements across diagnostic accuracy, stability, and test rationality without degrading information efficiency.

E6 also achieved the highest overall accuracy in the study (54.7%), surpassing even the instruction-aligned Llama-3.1-8B (E3: 53.5%). This supports the hypothesis that decoupling reasoning from dialogue generation at the adapter level works better than relying on a single model to handle both tasks at once. The additional VRAM overhead was negligible (≈50 MB for both adapters combined), confirming that dynamic LoRA routing is a practical enhancement even in resource-constrained deployments.

Chapter 7

Discussion

7.1 The Triumph of Structured Metacognition over Lexical Leakage

The most telling result from the v2 transition is what happened when we removed Lexical Leakage. In the initial system, the homogeneous Mistral-7B architecture achieved 44.0% accuracy, but there was a legitimate concern that some of that performance came from the Doctor "recognising" the Patient's embeddings through a shared manifold.

We expected accuracy to drop once we enforced strict heterogeneous engine isolation, since removing the implicit performance aids should, in theory, make the task harder. Instead, baseline accuracy **increased to 45.6%**. The LangGraph DCG and the forced metacognitive reflection node provide a reasoning scaffold strong enough to more than offset the loss of Lexical Leakage. By making the LLM populate a structured JSON differential *before* generating dialogue, the framework stabilises the clinical trajectory and suppresses the kind of iterative hallucinations that plagued the ad-hoc loop.

7.2 Domain Specialisation versus Instruction Alignment

The E2 and E3 results highlight an ongoing tension in medical AI. Fine-tuning on biomedical literature (JSL-MedMistral, E2) improves diagnostic vocabulary and pathological anchoring, but it tends to erode the broader logical reasoning and instruction-following abilities that multi-turn simulation demands. This *catastrophic forgetting* was already visible in the initial framework, where BioGPT (35%) and JSL-Med (32%) both fell short of the generalist Mistral-7B (44%) despite possessing greater biomedical knowledge [10].

A sufficiently large, highly instruction-aligned generalist model like Llama-3.1-8B (E3) appears better suited to the sequential, multi-conditional reasoning required for interactive clinical consultation. The implication is that for interactive clinical AI, *instruction alignment and logical deduction* matter as much as, if not more than, specialty medical recall. The ideal clinical agent likely needs both, which is exactly what the E6 Dynamic LoRA experiment set out to test.

7.3 Dynamic LoRA Routing: Resolving the Specialisation–Alignment Trade-off

E6 provides evidence that the specialisation–alignment trade-off is not inherent to LLM architecture but rather a consequence of monolithic weight modification. By deploying task-specific LoRA adapters that activate conditionally on the LangGraph node, the domain-expert JSL-MedMistral model reached 54.7% accuracy, surpassing both its unmodified baseline (E2: 50.2%) and the instruction-aligned Llama-3.1-8B (E3: 53.5%).

What this means in practice: the reasoning adapter concentrates the model's attention capacity on medical differential enumeration during the reflection phase, while the dialogue adapter maintains conversational coherence during patient interaction. Neither adapter catastrophically modifies the base model's pre-trained weights. The improvement in Diagnostic Stability (DS: 0.68 $\rightarrow$ 0.72) and Test

Rationality (TR: 0.71 $\rightarrow$ 0.76) confirms that these are not superficial accuracy gains but reflect genuinely improved reasoning trajectories.

7.4 The Safety Threat of Cognitive Biases

The E4 and E5 findings have serious implications for deploying open-source LLMs in clinical decision support. A model's context window is inherently a vector for recency bias: if an LLM sequentially processes multiple cardiac cases in an emergency room setting, its tendency to diagnose a subsequent, unrelated case as cardiac-related goes up because of the disproportionate attention weights assigned to recent tokens.

Without explicit architectural safeguards, such as enforced context resets between cases, constrained attention windowing, or metacognitive override prompts, LLMs will consistently replicate and may amplify human cognitive errors. The finding that confirmation bias produces a DS of $\approx$0.88 (high apparent stability) while simultaneously yielding the lowest accuracy ($\sim$31.4%) is particularly concerning: a model can appear *confidently consistent* while being systematically wrong. In a clinical setting, that kind of failure mode has direct patient safety consequences.

7.5 Implications for Clinical AI Evaluation Standards

A secondary contribution of this work is showing that binary diagnostic accuracy is an insufficient proxy for clinical AI safety. The process-oriented metrics introduced here—Diagnostic Stability, Test Rationality, and Information Efficiency—reveal behaviours that outcome accuracy entirely masks. A model that achieves 45.6% correct terminal diagnoses but exhibits $TR < 0.5$ (ordering invasive procedures before basic vitals) or $DS < 0.4$ (switching erratically between unrelated diagnoses) would be an unacceptable clinical risk, regardless of its accuracy number. Future evaluation standards for clinical LLMs should mandate process-level reporting alongside outcome accuracy [6].

7.6 Limitations

A key limitation of the current framework is that the Measurement Agent operates only in the text domain. Real-world diagnostics depend heavily on visual data; integrating Vision-Language Models (VLMs) to process raw ECG waveforms, chest radiographs, and histological slides is an important frontier that we have not yet addressed. All experiments were also conducted on the AgentClinic-MedQA subset, and generalisation to broader clinical datasets such as MIMIC-IV or NEJM case challenges would require separate investigation.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This thesis moved the evaluation of Large Language Models in clinical environments from static factual recall toward dynamic, interactive, and trajectory-aware simulation. By developing **AgentClinic v2**, we built a deterministic LangGraph state machine that runs entirely on local hardware via 4-bit NF4 double quantisation and heterogeneous multi-engine assignments, removing both the resource barrier and the Lexical Leakage vulnerability of the initial framework.

The latent metacognitive reflection node proved to be a powerful addition. Forcing an LLM to "think before it speaks" improved baseline accuracy from 44.0% (initial framework) to **45.6%** (AgentClinic v2), even after removing the implicit benefits of Lexical Leakage. Beyond accuracy, this thesis shows that clinical AI must be evaluated on its *process*—Diagnostic Stability, Test Rationality, and Information Efficiency—not just its terminal outcome.

Medical fine-tuning improves domain anchoring (E2: 50.2%), but instruction-aligned reasoning models perform even better for interactive tasks (E3: 53.5%). Dynamic LoRA adapter routing in E6 resolved this trade-off, achieving the highest accuracy in the study at **54.7%** by decoupling reasoning from dialogue generation at the adapter level. At the same time, open-source models remain vulnerable to cognitive distortions: recency bias causes erratic hypothesis thrashing (E4: 38.1%), and

confirmation bias leads to premature diagnostic closure (E5: 31.4%). These results make clear that cognitive robustness must be treated as a primary design criterion in any clinical LLM deployment.

8.2 Future Work: Advanced Fine-Tuning and Post-Training Alignment

While this thesis establishes a framework for evaluating multi-turn clinical reasoning, the next step is to actively mitigate the failure modes we identified. Future work will use AgentClinic v2 not just as an evaluator, but as an environment for advanced post-training alignment.

8.2.1 Process Reward Modelling via Direct Preference Optimisation

Current medical LLMs are typically trained via Supervised Fine-Tuning (SFT) on static QA pairs, optimised to predict the correct terminal diagnosis but with no understanding of the *trajectory* needed to reach it safely. We propose applying Direct Preference Optimisation (DPO) to full clinical consultation trajectories. Instead of rewarding only the correct final outcome, a Process Reward Model would evaluate two clinical paths: Path A (premature hypothesis commitment with low Test Rationality) versus Path B (systematic basic-before-advanced investigation with high Diagnostic Stability). By penalising Path A and rewarding Path B, DPO would adjust the LLM's behavioural policy to prefer methodical, guideline-concordant reasoning.

8.2.2 Reinforcement Learning from AI Feedback via Self-Play

Generating expert, human-annotated clinical dialogues for fine-tuning is expensive and difficult to scale. A natural alternative is to use the AgentClinic framework as a Reinforcement Learning (RL) environment. By allowing heterogeneous agents

to simulate thousands of cases through self-play, synthetic training data can be generated automatically. The `ClinicalTrajectoryEvaluator` developed in this thesis would serve as the programmatic reward function: trajectories scoring highly on DS and TR would be archived, enabling iterative fine-tuning on the model's own successful reasoning paths. This would create a self-improving clinical agent without the overhead of human annotation.

8.2.3 Extending Dynamic LoRA to Bias Resilience

The dynamic LoRA routing in E6 showed clear gains in unbiased settings. A natural extension is to train *debiasing adapters* aimed at counteracting the failure modes observed in E4 and E5. A recency-debiasing adapter could be trained on trajectory pairs where the model successfully resisted anchoring to injected case memories, while a confirmation-debiasing adapter could be optimised to maintain high Test Rationality even under strong prior hypotheses. Combining these with the existing reasoning and dialogue adapters would create a four-adapter routing manifold, selected dynamically by the LangGraph orchestrator based on both the active node and detected bias risk.

8.2.4 Multimodal Simulation Integration

The current `measurement_node` operates only in the text domain. Real-world diagnostics rely heavily on visual pattern recognition. Future versions of AgentClinic should integrate Vision-Language Models (VLMs) such as LLaVA-Med or Med-Gemini, enabling the Measurement Agent to return raw radiographic images, ECG waveforms, and pathology slides for direct interpretation by the Doctor agent, closing the remaining gap toward end-to-end clinical simulation.

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